EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Hydrogen; Isotopes of Hydrogen; Types of Hydrogen

1. Ortho and Para hydrogen differ :-

(A) In the number of neutrons (B) In the chemical properties

(C) In the physical properties (D) In both chemical & physical properties

CHY153

Preparation of Hydrogen (Laboratory preparation and Commercial production of H_2)

2. Hydrogen from HCl can be prepared by:

(A) Cu (B) P (C) Mg (D) Hg

CHY081

Reactions of Hydrogen (Reactions with Halogen, O_2, N_2, Metals, Metal Ions, and Metal Oxides, Organic Compounds)

3. Reaction between following pairs will produce hydrogen except :

(A) Cu + HCl (B) Fe + H_2O (g) (C) Mg + H_2O (hot) (D) Na + Alcohol

CHY084

4. Which of the following statements is most applicable to hydrogen?

(A) It can act as a reducing agent only

(B) It can act as an oxidising agent only

(C) It can act as both as oxidising and reducing agents

(D) It can act neither as an oxidising nor as a reducing agent

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CHY085

5. The metal which displaces hydrogen from a boiling caustic soda solution is :

(A) Mg (B) Fe (C) As (D) Zn

CHY082

Uses of Dihydrogen, Dihydrogen as a fuel

6. The colour of hydrogen is :

(A) black (B) yellow (C) orange (D) colourless

CHY155

Hydrides

7. Which of the following is a interstitial hydride?

(A) $TiH_{1.5-1.8}$ (B) B_2H_6 (C) LiH (D) H_2S

CHY086

8. Which of the following represents a pair of covalent hydrides ?

(A) CsH, AlH_3 (B) KH, NaH (C) H_2S, HF (D) $VH_{0.56}$, NH_3

CHY087

9. The hydride ion H^- is a stronger base than its hydroxide ion OH^-. Which of the following reactions will occur if sodium hydride (NaH) is dissolved in water?

(A) $H^-(aq) + H_2O \longrightarrow H_3O^-(aq)$ (B) $H^-(aq) + H_2O(l) \longrightarrow OH^-(aq) + H_2(g)$

(C) $H^-(aq) + H_2O(l) \longrightarrow$ No reaction (D) None of these.

CHY088

For More Join: @IITJEE_Advanced

10. NaH is an example of :
(A) Electron-rich hydride (B) Saline hydride
(C) Molecular hydride (D) None of these

CHY089

Heavy Water

11. Select the correct statement for heavy water.
(A) It is less denser than common water. (B) It is an oxide of deuterium.
(C) It has a heavy or bad taste. (D) It has a heavier isotope of oxygen.

CHY092

12. Heavy water is used in nuclear reactors as :
(A) source of a particles.
(B) slowing down the speed of high energy neutrons.
(C) transporting heat of the reactor.
(D) heating purposes.

CHY093

Water and Hardness of Water

13. High boiling point of water as compared to "H_2S" is due to :-
(A) Its high specific heat (B) Hydrogen bonding
(C) High dielectric constant (D) Low dissociation constant

CHY091

14. Hard water when passed through ion exchange resin containing, RCOOH groups, becomes free from :
(A) Cl^- ions (B) SO_4^{2-} ions (C) H_3O^+ ions (D) Ca^{2+} ions

CHY094

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15. What is false about ice?
(A) It has open cage like structure.
(B) It has less density than water.
(C) Each O atom is surrounded by four H atoms.
(D) Each O atom has four H-bonds around it.

CHY095

16. Temporary hardness may be removed from water by adding :
(A) $Ca(OH)_2$ (B) $CaCO_3$ (C) $CaSO_4$ (D) HCl

CHY097

17. Both temporary and permanent hardness is removed on boiling water with :-
(A) $Ca(OH)_2$ (B) Na_2CO_3 (C) $CaCO_3$ (D) CaO

CHY157

18. The temporary hardness of a water sample is due to compound X. Boiling this sample converts X to compound Y. X and Y respectively, are :
(A) $Ca(HCO_3)_2$ and CaO (B) $Mg(HCO_3)_2$ and $MgCO_3$
(C) $Mg(HCO_3)_2$ and $Mg(OH)_2$ (D) $Ca(HCO_3)_2$ and $Ca(OH)_2$

CHY158

19. Which of the following can effectively remove all types of hardness of water
(A) Soap (B) Washing soda (C) Slaked lime (D) Sodium chloride

CHY160

For More Join: @IITJEE_Advanced

Hydrogen Peroxide, Preparation and Properties

20. The bleaching properties of H_2O_2 are due to its :-
(A) Reducing properties (B) Oxidising properties
(C) Unstable nature (D) Acidic nature

CHY099

21. H_2O_2 is always stored in black bottles because :-
(A) It is highly unstable
(B) Its enthalpy of decomposition is high
(C) It undergoes auto-oxidation on prolonged standing
(D) None of these

CHY100

22. In acidic medium, H_2O_2 acts as a reducing agent in its reaction with :
(A) $FeSO_4$ (B) $KMnO_4$ (C) K_2MnO_4 (D) $K_4[Fe(CN)_6]$

CHY104

23. In basic medium, H_2O_2 acts as an oxidising agent in its reactions with :
(A) $Cr_2(SO_4)_3$ (B) Ag_2O (C) $K_3[Fe(CN)_6]$ (D) $K_2Cr_2O_7$

CHY105

24. Which of the following compounds turns white on treatment with H_2O_2 ?
(A) HgS (B) PbS (C) NiS (D) CuS

CHY106

25. On an industrial scale, H_2O_2 is prepared by auto-oxidation of :
(A) 2-Ethylanthraquinol. (B) 2-Ethylanthraquinone
(C) 1-Ethylanthraquinol. (D) 1-Ethylanthraquinone.

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CHY108

26. In which of the following reaction hydrogen peroxide is a reducing agent
(A) $2FeCl_2 + 2HCl + H_2O_2 \rightarrow 2FeCl_3 + 2H_2O$
(B) $Cl_2 + H_2O_2 \rightarrow 2HCl + O_2$
(C) $2HI + H_2O_2 \rightarrow 2H_2O + I_2$
(D) $H_2SO_3 + H_2O_2 \rightarrow H_2SO_4 + H_2O$

CHY163

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Hydrogen; Isotopes of Hydrogen; Types of Hydrogen

1. Which of the following statements is/are correct :
(A) Atomic hydrogen is obtained by passing hydrogen gas through an electric arc
(B) 30% (w/v) or 100V H_2O_2 solution is not called per hydrol.
(C) Finely divided palladium absorbs large volume of hydrogen gas.
(D) Ortho and para hydrogen have same physical properties.

CHY164

Preparation of Hydrogen (Laboratory preparation and Commercial production of H_2)

2. Which of the following process can be used for preparation of H_2 gas?
(A) Dissolving LiH in water
(B) Reaction of Al with NaOH solution
(C) Reaction of Zn with dilute H_2SO_4
(D) Electrolysis of H_2O in the presenece of KOH

CHY111

3. Which of the following pair of compound(s) can be used for producing hydrogen gas?
(A) Be + dil. H_2SO_4 (B) Be + NaOH (C) Al + HCl (g) (D) Al + NaOH

CHY165

Reactions of Hydrogen (Reactions with Halogen, O_2, N_2, Metals, Metal Ions, and Metal Oxides, Organic Compounds)

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4. Which of the following reaction form H_2 gas
(A) Na $\xrightarrow{H_2O}$ (B) Zn $\xrightarrow{\text{dil. } H_2SO_4}$ (C) Zn $\xrightarrow{\text{conc. HCl}}$ (D) Zn $\xrightarrow{\text{conc. } H_2SO_4}$

CHY166

Hydrides

5. Which of the following properties of metallic hydrides are similar to those of their parent metal ?
(A) Hardness
(B) Metallic lustre
(C) Electrical conductivity
(D) Magnetic property

CHY112

6. Which of the following statements is correct regarding the conversion of metal into metallic hydride ?
(A) The density lattic expands though the inclusion of hydride.
(B) The crystel lattice expands through the inclusion of hydride.
(C) A solid piece of a metal turns into powdered hydride
(D) None of these.

CHY114

Water and Hardness of Water

7. Some of the properties of water are described below. Which of them is/are not correct ?
(A) Water is known to be a universal solvent.
(B) Hydrogen bonding is present to a large extent in liquid water.
(C) There is no hydrogen bonding in the frozen state of water.
(D) Frozen water is heavier than liquid water.

CHY167

For More Join: @IITJEE_Advanced

COMPREHENSION TYPE QUESTIONS

Uses of Dihydrogen, Dihydrogen as a fuel

Passage for Q.8 to Q.10

Hydrogen accounts for approximately 75% of the mass of the universe. Hydrogen serves as the nuclear fuel of our Sun and other stars, and these are mainly composed of hydrogen.

Hydrogen has three isotopes : hydrogen or protium ($_1H^1$), deuterium or heavy hydrogen (D or $_1H^2$), tritium (T or $_1H^3$).

8. Which of the following is radioactive in nature ?

(A) hydrogen only
(B) deuterium only
(C) tritium only
(D) deuterium and tritium

CHY115

9. Hydrogen, H_2, is very less abundant in the atmosphere due to -

(A) inflammable nature of H_2
(B) weak earth's gravity which is not able to hold light H_2 molecules
(C) diatomic nature of hydrogen
(D) very rapid reaction between hydrogen and atmospheric oxygen

CHY116

10. Liquid H_2 has been used as rocket fuel as

(A) its reaction with oxygen is highly exothermic
(B) it occupies small space
(C) it has high thrust
(D) All of the above

Rankers Academy JEE

CHY117

For More Join: @IITJEE_Advanced

EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

Hydrogen; Isotopes of Hydrogen; Types of Hydrogen

1. Hydrogen has three isotopes (A), (B) and (C).
If the number of neutron(s) in (A), (B) and (C) respectively, are (x), (y) and (z), the multiplication of (y) and (z) is :

CHY119

2. The number of protons that can be accepted by hydrazine is_____.

CHY121

3. Percentage of para hydrogen at low temperature

CHY169

Water and Hardness of Water

4. How many water molecules are associated with 'Epsom salt'?

CHY171

Part-2 : JEE Advanced (Decimal type)

Reactions of Hydrogen (Reactions with Halogen, O_2, N_2, Metals, Metal Ions, and Metal Oxides, Organic Compounds)

5. Hydrogen acts as an oxidising agent in the how many reaction with :
(i) Bromine (ii) Calcium (iii) Nitrogen (iv) Sulphur

CHY172

6. Find out the value of $\frac{x}{2}$ in ion $[H_xO_4]^+$:

CHY122

Rankers Academy JEE

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

Hydrogen; Isotopes of Hydrogen; Types of Hydrogen

1. Given below are two statements, one is labelled as **Assertion A** and the other is labelled as **Reason R**.
Assertion A : Isotopes of hydrogen have almost same chemical properties, but difference in their rates of reaction.
Reason R : Isotopes of hydrogen have different enthalpy of bond dissociation.
In the light of the above statements, choose the most appropriate answer from the options given below: **[JEE (Main) 2023]**
(1) Both A and R are correct but R is NOT the correct explanation of A
(2) Both A and R are correct and R is the correct explanation of A
(3) A is not correct but R is correct
(4) A is correct but R is not correct

CHY141

2. The highest industrial consumption of molecular hydrogen is to produce compounds of element: **[JEE (Main) 2022]**
(1) Carbon (2) Nitrogen (3) Oxygen (4) Chlorine

CHY130

3. Which one of the following statements is incorrect ? **[JEE (Main) 2021]**
(1) Atomic hydrogen is produced when H_2 molecules at a high temperature are irradiated with UV radiation.
(2) At around 2000 K, the dissociation of dihydrogen into its atoms is nearly 8.1%.
(3) Bond dissociation enthalpy of H_2 is highest among diatomic gaseous molecules which contain a single bond.
(4) Dihydrogen is produced on reacting zinc with HCl as well as NaOH(aq).

CHY125

Bankers Academy JEE

Preparation of Hydrogen (Laboratory preparation and Commercial production of H_2)

4. Given below are two statements:
Statement I : Nickel is being used as the catalyst for producing syn gas and edible fats.
Statement II : Silicon forms both electron rich and electron deficient hydrides.
In the light of the above statements, choose the most appropriate answer from the options given below: **[JEE (Main) 2023]**
(1) Both the statements I and II are correct
(2) Statement I is incorrect but statement II is correct
(3) Both the statements I and II are incorrect
(4) Statement I is correct but statement II is incorrect

CHY136

5. During water-gas shift reaction **[JEE (Main) 2023]**
(1) carbon monoxide is oxidized to carbon dioxide.
(2) carbon is oxidized to carbon monoxide.
(3) carbon dioxide is reduced to carbon monoxide.
(4) water is evaporated in presence of catalyst.

CHY142

For More Join: @IITJEE_Advanced

6. Which of the following has least tendency to liberate H_2 from mineral acids? **[JEE (Main) 2022]**

(1) Cu (2) Mn (3) Ni (4) Zn

CHY129

7. The reaction of zinc with excess of aqueous alkali, evolves hydrogen gas and gives:

[JEE (Main) 2022]

(1) $Zn(OH)_2$ (2) ZnO (3) $[Zn(OH)_4]^{2-}$ (4) $[ZnO_2]^{2-}$

CHY131

Reactions of Hydrogen (Reactions with Halogen, O_2, N_2, Metals, Metal Ions, and Metal Oxides, Organic Compounds)

8. High purity (>99.95%) dihydrogen is obtained by **[JEE (Main) 2022]**

(1) reaction of zinc with aqueous alkali.

(2) electrolysis of acidified water using platinum electrodes.

(3) electrolysis of warm aqueous barium hydroxide solution between nickel electrodes.

(4) reaction of zinc with dilute acid.

CHY127

Uses of Dihydrogen, Dihydrogen as a fuel

9. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**

Assertion A: Hydrogen is an environment friendly fuel.

Reason R: Atomic number of hydrogens is 1 and it is a very light element.

In the light of the above statements, choose the correct answer from the options given below

[JEE (Main) 2023]

(1) A is true but R is false

(2) Both A and R are true but R is NOT the correct explanation of A

(3) A is false but R is true

(4) Both A and R are true and R is the correct explanation of A

CHY139

Kankers Academy JEE

Water and Hardness of Water

10. Decreasing order of the hydrogen bonding in following forms of water is correctly represented by

A. Liquid water

B. Ice

C. Impure water **[JEE (Main) 2023]**

(1) A = B > C (2) B > A > C (3) C > B > A (4) A > B > C

CHY133

Hydrogen Peroxide, Preparation and Properties

11. Given below are two statements :

Statement I : H_2O_2 is used in the synthesis of Cephalosporin

Statement II : H_2O_2 is used for the restoration of aerobic conditions to sewage wastes.

In the light of the above statements, choose the **most appropriate** answer from the options given below : **[JEE (Main) 2023]**

(1) Both **Statement I** and **Statement II** are correct

(2) **Statement I** is incorrect but **Statement II** is correct

(3) **Statement I** is correct but **Statement II** is incorrect

(4) Both **Statement I** and **Statement II** are incorrect

CHY138

For More Join: @IITJEE_Advanced

12. Which of the following can reduce decomposition of H_2O_2 on exposure to light **[JEE (Main) 2023]**

(1) Alkali (2) Urea (3) Dust (4) Glass containers

CHY140

13. Match List I with List II

	List I		List II
A.	Cobalt catalyst	I.	(H_2 + Cl_2) production
B.	Syngas	II.	Water gas production
C.	Nickel catalyst	III.	Coal gasification
D.	Brine solution	IV.	Methanol production

Choose the correct answer from the options given below : **[JEE (Main) 2023]**

(1) A-IV, B-I, C-II, D-III (2) A-IV, B-III, C-I, D-II

(3) A-II, B-III, C-IV, D-I (4) A-IV, B-III, C-II, D-I

CHY135

14. Given below are two statements: **[JEE (Main) 2022]**

Statement I : Hydrogen peroxide can act as an oxidizing agent in both acidic and basic conditions.

Statement II : Density of hydrogen peroxide at 298 K is lower than that of D_2O.

In the light of the above statements. Choose the ***correct*** answer from the options.

(1) Both statement I and Statement II are true

(2) Both statement I and Statement II are false

(3) Statement I is true but Statement II is false

(4) Statement I is false but Statement II is true

CHY126

Rankers Academy JEE

15. The metal salts formed during softening of hardwater using Clark's method are:

[JEE (Main) 2022]

(1) $Ca(OH)_2$ and $Mg(OH)_2$ (2) $CaCO_3$ and $Mg(OH)_2$

(3) $Ca(OH)_2$ and $MgCO_3$ (4) $CaCO_3$ and $MgCO_3$

CHY128

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

Water and Hardness of Water

1. The reagent(s) used for softening the temporary hardness of water is(are) – **[JEE (Advanced) 2010]**

(A) $Ca_3(PO_4)_2$ (B) $Ca(OH)_2$ (C) Na_2CO_3 (D) NaOCl

CHY073

2. Polyphosphates are used as water softening agent because they :- **[IIT-2002]**

(A) Form soluble complexes with anionic species

(B) Precipitate anionic species

(C) Form soluble complexes with cationic species

(D) Precipitate cationic species.

CHY072

Hydrogen Peroxide, Preparation and Properties

3. Hydrogen peroxide in its reaction with KIO_4 and NH_2OH respectively, is acting as a **[JEE (Advanced) 2014]**

(A) reducing agent, oxidising agent (B) reducing agent, reducing agent

(C) oxidising agent, oxidising agent (D) oxidising agent, reducing agent

CHY074

Rankers Academy JEE

For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	C	A	C	D	D	A	C	B	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	B	B	D	D	A	B	C	B	B
Que.	21	22	23	24	25	26				
Ans.	C	B	A	B	A	B				

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	AC	ABCD	ABCD	ABC	ABCD	ABC	CD	C	B	D

EXERCISE - S

Que.	1	2	3	4	5	6
Ans.	2	2	100	7	1	4.5

Rankers Academy JEE

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	2	2	4	1	1	4	3	2	2
Que.	11	12	13	14	15					
Ans.	1	2	4	3	2					

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3
Ans.	BC	C	A

For More Join: @IITJEE_Advanced